Anaesth Intensive Care 2005; **33:** 41-53

Use of Intrathecal Neostigmine as an Adjunct to Other Spinal Medications in Perioperative and Peripartum Analgesia: a Meta-analysis

K. M. HO*, H. ISMAIL†, K. C. LEE‡, R. BRANCH§

Department of Anaesthesia and Intensive Care, North Shore Hospital, Auckland, New Zealand

SUMMARY

Intrathecal neostigmine has been used as an adjunct to intrathecal local anaesthetic or opioid to prolong regional analgesia and improve haemodynamic stability, with variable results. This meta-analysis aims to evaluate the effectiveness and side-effects of intrathecal neostigmine in the perioperative and peripartum settings. The literature search was based on Cochrane Controlled Trials Register, EMBASE and MEDLINE (from 1966 to 14 November 2003) databases. Volunteer and animal studies were excluded. We identified 26 studies and 19 were considered suitable for detailed data extraction. Intrathecal neostigmine increased the incidence of nausea and vomiting (OR 5.0, 95% CI: 3.4 to 7.3; P<0.00001), bradycardia requiring intravenous atropine (OR 2.7, 95% CI: 1.4 to 5.4; P=0.005), and anxiety, agitation, or restlessness (OR 10.3, 95% CI: 3.7 to 28.9; P=0.00001). It improved the overall 24 hour VAS score (–1.4 VAS pain score, 95% CI: –1.7 to –1.2, P<0.00001), delayed the time of first request for rescue analgesia (168 min, 95% CI: 125 to 211; P<0.00001), and reduced the total number of rescue injections of nonsteroidal anti-inflammatory drug within the first 24 hours (–0.8, 95% CI: –1.1 to –0.4; P=0.00001). It did not affect the duration of motor blockade (3.5 min, 95% CI: –1.5 to 8.6; P=0.17) or the total amount of ephedrine required (–0.4 mg, 95% CI: –1.5 to 0.7; P=0.5). Adding intrathecal neostigmine to other spinal medications improves perioperative and peripartum analgesia marginally when compared with placebo. It is associated with significant side-effects and the disadvantages outweigh the minor improvement in analgesia achieved.

Key Words: ANALGESIA, REGIONAL: neostigmine, intrathecal

Intrathecal neostigmine inhibits the breakdown of an endogenous spinal neurotransmitter, acetylcholine. It has been shown to produce analgesia on its own[1] and to reduce the risk of hypotension induced by spinal local anaesthetic in animal studies[2,3]. Both animal and human studies show that intrathecal neostigmine does not affect spinal cord blood flow and does not induce any significant toxic effect in the spinal cord[4-6]. However, significant systemic side-effects of intrathecal neostigmine, such as nausea and vomiting, motor blockade, sedation, and hypertension, have been reported, especially with doses greater than 200 μg[5]. A recent systematic review included a small number of randomized controlled trials and could not quantify the potential clinical benefits and side-effects of intrathecal neostigmine[7]. A number of clinical studies, using lower doses of intrathecal neostigmine, have been published since the last review, with variable results regarding its analgesic effect and side-effects. We aimed to re-evaluate the use of intrathecal neostigmine as an adjunct in spinal anaesthesia for the prevention and treatment of acute perioperative and peripartum pain. In particular, we wanted to quantify the potential benefits and side-effects of intrathecal neostigmine when used as an adjunct to other intrathecal medications in spinal anaesthesia or analgesia.

MATERIALS AND METHODS

A literature search was based on Cochrane Controlled Trials Register (2003 issue 4), MEDLINE and EMBASE (1996 to 14 November 2003) databases. Only randomized controlled clinical trials or

* M.P.H., M.R.C.P., F.A.N.Z.C.A., F.J.F.I.C.M., Consultant Anaesthetist/ Intensivist, Department of Anaesthesia and Intensive Care, North Shore Hospital, Auckland, New Zealand.

† M.D., F.R.C.A., F.A.N.Z.C.A., Consultant Anaesthetist, Department of Anaesthesia and Intensive Care, North Shore Hospital, Auckland, New Zealand.

‡ B.M.B.S., F.A.N.Z.C.A., Consultant Anaesthetist, Department of Anaesthesia and Intensive Care, North Shore Hospital, Auckland, New Zealand.

§ M.B.Ch.B., Anaesthetic Registrar, Department of Anaesthesia and Intensive Care, North Shore Hospital, Auckland, New Zealand.

Address for reprints: Dr K. M. Ho, Intensive Care Unit, Royal Perth Hospital, Perth, W.A. 6000.

Accepted for publication on October 26, 2004.

quasi-randomized controlled clinical trials in the perioperative and peripartum settings were included in this meta-analysis. Studies with more than one active component were included if both intervention and control groups were exposed to the other active component. Clinical trials that compared intrathecal neostigmine with another active intervention such as intrathecal morphine but without a separate placebo control group in the same trial were excluded. Trials that involved significant dissimilar co-interventions, such as one group of patients receiving intrathecal local anaesthetic with intrathecal neostigmine and the other group only placebo, were excluded. Volunteer and animal studies were also excluded.

During the electronic database search, the following exploded MeSH terms were used: "neostigmine" and "intrathecal", "subarachnoid", or "spinal". The reference lists of related reviews and identified original articles were checked for relevant trials. Finally, the websites of International Network of Agencies of Health Technology Assessment and International Society of Technology Assessment in Health Care were searched. If necessary, the authors of the identified trials were contacted to obtain additional vital information and unpublished data that were considered important in the data analysis. Studies published in other languages other than English were translated and included in this meta-analysis.

Two independent reviewers examined the titles and the abstracts of all identified trials to confirm they fulfilled the above-defined inclusion criteria. They examined and recorded the trial characteristics and outcomes independently, using a pre-designed article abstraction form. This abstraction form was used to record information regarding the quality of the trial, such as allocation concealment, randomization method, blinding of treatment, inclusion and exclusion criteria. The grading of allocation concealment was based on the Cochrane approach, i.e. adequate or uncertain or clearly inadequate. Any disagreements between the two independent reviewers were resolved by consensus or by a third reviewer if there was no consensus. Articles that fulfilled the inclusion criteria were analysed further for data extraction.

The incidences of nausea and vomiting, bradycardia requiring the use of intravenous atropine, involuntary defaecation, and anxiety, agitation, or restlessness were recorded as categorical variables. The time to request first rescue analgesia, the duration of motor blockade till recovery to Bromage grade 2, the total amount of intravenous ephedrine required, the overall 24-hour visual analogue scale (VAS) pain score, and the total number of intramuscular nonsteroidal anti-inflammatory drug (NSAID) injections in 24 hours, were considered continuous variables and mean and standard deviations extracted and recorded in the data abstraction form. Two reviewers extracted the data independently and the results were checked for consistency. Any duplicate publications were combined to represent one single trial. Data were checked and entered into Review Manager (version 4.1) database by two independent reviewers and the results checked.

Statistical Analysis

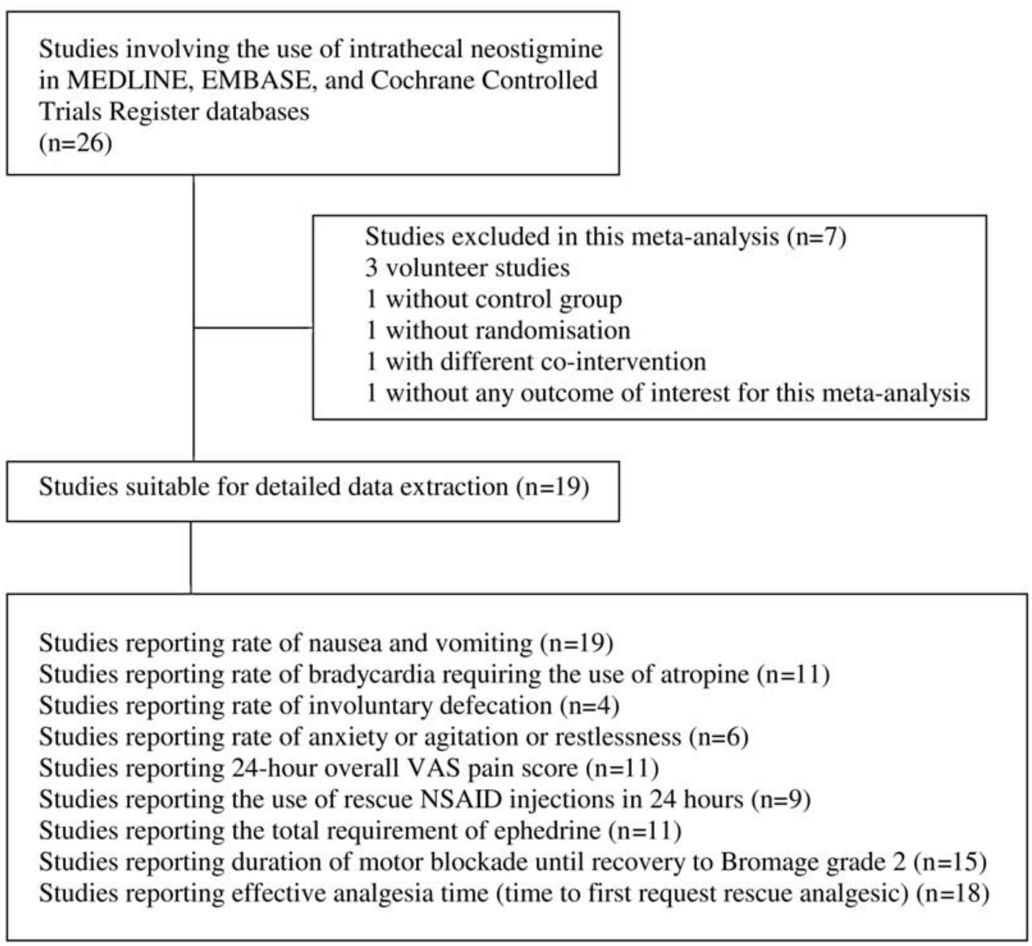
The proportion of patients with different side-effects such as nausea and vomiting, bradycardia requiring the use of intravenous atropine, involuntary defaecation, and anxiety, agitation, or restlessness were reported as odds ratios (OR) with 95% confidence intervals (95% CI), using random effect model. The continuous outcome variables were reported as weighted mean difference (WMD) in the original unit of measurement with 95% CI, using random effect model. If only the median and range were reported in the original studies, but the data were nearly normally distributed, the median would be assigned as the mean and the standard deviation estimated as (0.95×range)/4. The 24-hour VAS pain score was assessed by asking the patient the overall pain control over 24 hours and this ranged from 0 to 10 in most studies. The VAS pain data was reported on a scale of 0 to 5 in one study and the data were converted to a scale of 10. The presence of heterogeneity between trials was assessed by chi-squared test. The extent of inconsistency[8] among results of the trials was assessed by statistics I^2. Sensitivity analysis, such as excluding poor quality trials (trials with clearly inadequate allocation concealment or blinding of randomization) was conducted to test the robustness of the results. Multiple parallel comparisons in the same study were labelled with the same study identity name followed by either a, b, c, or d in the Forest plots. In order to facilitate the interpretation of any dose response relationship in the effectiveness and side-effect profiles of intrathecal neostigmine, studies were ranked in all the Forest plots according to the dose of intrathecal neostigmine used, with the study using the highest dose of intrathecal neostigmine at the top of the plots. Publication bias was assessed by funnel plot.

RESULTS

We identified 26 human studies involving the use of intrathecal neostigmine and 19 were considered suitable for detailed data extraction. Among the seven studies that were excluded, three were volunteer studies[9-11], one study did not use a control group[12], one study used a different co-intervention between the neostigmine group and control group[13], one study did not assess any outcome of interest included in this meta-analysis[14], and one study did not use randomization in the study (Figure 1)[15]. There was no difference in the assessment of studies, suitability for inclusion and only minor differences in data extraction between the two investigators. The minor differences in data extraction were resolved after carefully examining the studies together.

Twelve studies compared spinal local anaesthetic with intrathecal neostigmine or placebo. Two studies involved concurrent intrathecal clonidine, opioid and local anaesthetic in both the neostigmine and control groups[16,17]. One study involved general anaesthesia and intrathecal morphine in both the neostigmine and control groups[18]. Two studies used concurrent intrathecal opioid and spinal local anaesthetic in both the neostigmine and control groups[19,20]. One study involved concurrent intrathecal clonidine and local anaesthetic in both the neostigmine and control groups[21]. One study made three parallel comparisons involving intravenous placebo, intravenous fentanyl or intravenous ketamine in both the neostigmine and control groups[22]. Across all studies, ten different doses of intrathecal neostigmine, ranging from 1 to 500 μg, were used. Seven studies assessed more than one dose of intrathecal neostigmine[18,19,23-27]. In addition, three other studies involved more than one active treatment group, with different concurrent treatments in different active treatment groups and a comparable control group for each different active treatment group[20-22].

Nine studies involved patients undergoing gynaecologic procedures[18-20,22,27-31], two studies involved parturients undergoing elective caesarean delivery[21,32], two studies assessed intrathecal neostigmine for labour analgesia[16,17], three studies involved patients undergoing lower limb orthopaedic operations[25,33,34], and three studies involved patients undergoing



VAS=visual analog scale; NSAID=nonsteroidal anti-inflammatory drug

FIGURE 1: Flow chart showing study inclusion and exclusion in this meta-analysis.

general surgical operations in the lower abdomen or perineum[23-25]. All except one study involved patients with mean age less than 60 years old. Allocation concealment was clearly adequate in 10 studies. Eighteen studies were in English and one in Turkish. All studies used ephedrine as the vasopressor for hypotension when a predefined end-point was reached and fluid hydration was a bolus before spinal anaesthesia, with or without a fixed infusion rate as maintenance. The details of all included studies are described in Appendix 1.

The results of all the categorical outcome variables were homogenous. Intrathecal neostigmine significantly increased the incidence of nausea and vomiting (OR 5.0, 95% CI: 3.4 to 7.3; $P<0.00001$, $I^2=16\%$), bradycardia requiring intravenous atropine (OR 2.7, 95% CI: 1.4 to 5.4; $P=0.005$, $I^2=0\%$), and anxiety, agitation, or restlessness (OR 10.3, 95% CI: 3.7 to 28.9; $P=0.00001$, $I^2=0\%$). There was a trend toward an increased rate of faecal incontinence after intrathecal neostigmine (OR 3.0, 95% CI: 0.9 to 10.5, $P=0.08$, $I^2=0\%$). No obvious dose dependent relationship in the categorical outcome variables was observed when the trend of mean odds ratios was assessed in each forest plot (Figures 2-5).

Adding intrathecal neostigmine to other spinal medications improved the overall 24-hour VAS score (–1.4 VAS pain score, 95% CI: –1.7 to –1.2, $P<0.00001$, $I^2=0\%$), delayed the time of first request for rescue analgesia (168 min, 95% CI: 125 to 211; $P<0.00001$, $I^2=97\%$), but only slightly reduced the total number of rescue NSAID injections required within the first 24 hours (–0.8, 95%CI: –1.1 to –0.4; $P=0.00001$, $I^2=90\%$) (Figures 6-8). Intrathecal neostigmine did not increase the duration of motor blockade (3.5 min, 95% CI: –1.5 to 8.6; $P=0.17$, $I^2=11\%$) and did not reduce the total amount of ephedrine required to maintain haemodynamic stability intraoperatively (–0.4 mg, 95%CI: –1.5 to 0.7; $P=0.5$, $I^2=0\%$).

Subgroup analysis of studies involving only parturients receiving intrathecal neostigmine for elective

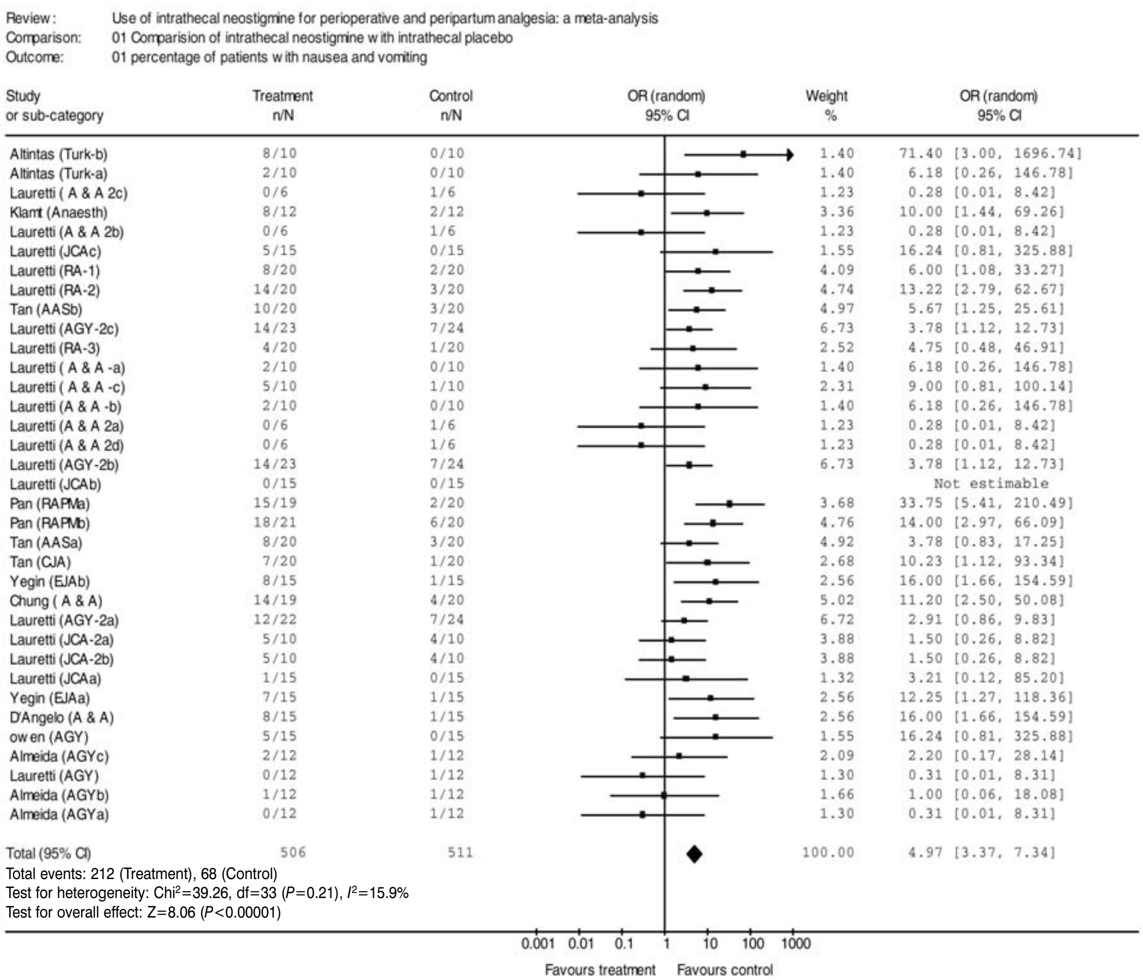


FIGURE 2: Forest plot showing the incidence of nausea and vomiting after treatment with intrathecal neostigmine.

Review: Use of intrathecal neostigmine for perioperative and peripartum analgesia: a meta-analysis
Comparison: 01 Comparision of intrathecal neostigmine w ith intrathecal placebo
Outcome: 02 percentage of patients w ith bradycardia requiring intravenous atropine

Study or sub-category	Treatment n/N	Control n/N	Weight %	OR (random) 95% CI
Yegin (EJAa)	1/15	0/15	4.36	3.21 [0.12, 85.20]
Yegin (EJAb)	1/15	0/15	4.36	3.21 [0.12, 85.20]
Lauretti (A & A 2c)	2/6	0/6	4.40	7.22 [0.28, 189.19]
Lauretti (A & A 2b)	0/6	0/6		Not estimable
Lauretti (JCAc)	2/15	0/15	4.81	5.74 [0.25, 130.37]
Lauretti (RA-1)	3/20	2/20	12.89	1.59 [0.24, 10.70]
Lauretti (RA-2)	1/20	0/20	4.42	3.15 [0.12, 82.16]
Lauretti (AGY-2c)	2/23	0/24	4.91	5.70 [0.26, 125.36]
Lauretti (RA-3)	2/20	1/20	7.60	2.11 [0.18, 25.35]
Lauretti (A & A -a)	2/10	0/10	4.68	6.18 [0.26, 146.78]
Lauretti (A & A -c)	3/10	0/10	4.86	9.80 [0.44, 219.25]
Lauretti (A & A -b)	1/10	0/10	4.26	3.32 [0.12, 91.60]
Lauretti (A & A 2a)	1/6	0/6	4.07	3.55 [0.12, 105.82]
Lauretti (A & A 2d)	0/6	0/6		Not estimable
Lauretti (AGY-2b)	3/23	0/24	5.14	8.37 [0.41, 171.55]
Lauretti (JCAb)	0/15	0/15		Not estimable
Lauretti (AGY-2a)	1/22	0/24	4.44	3.42 [0.13, 88.40]
Lauretti (JCA-2a)	1/10	3/10	7.70	0.26 [0.02, 3.06]
Lauretti (JCA-2b)	2/10	1/10	7.04	2.25 [0.17, 29.77]
Lauretti (JCAa)	1/15	0/15	4.36	3.21 [0.12, 85.20]
D'Angelo (A & A)	1/15	1/15	5.70	1.00 [0.06, 17.62]
ow en (AGY)	0/15	0/15		Not estimable
Total (95% CI)	307	311	100.00	2.70 [1.36, 5.35]

Total events: 30 (Treatment), 8 (Control)
Test for heterogeneity: Chi^2=6.64, df=17 (P=0.99), I^2=0%
Test for overall effect: Z=2.84 (P=0.005)

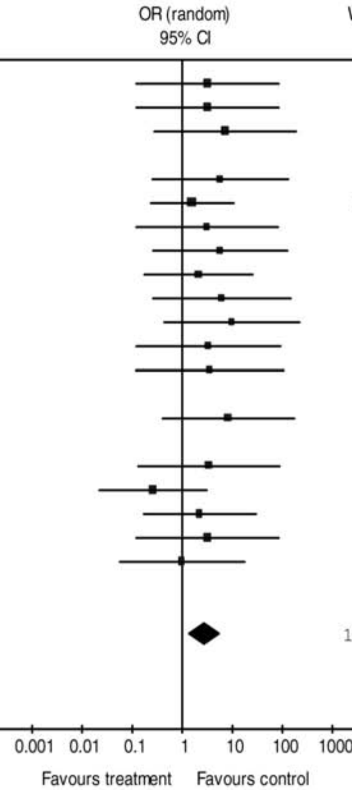


FIGURE 3: Forest plot showing the incidence bradycardia requiring atropine after treatment with intrathecal nesotigmine.

Review: Use of intrathecal neostigmine for perioperative and peripartum analgesia: a meta-analysis
Comparison: 01 Comparision of intrathecal neostigmine w ith intrathecal placebo
Outcome: 03 percentage of patients w ith faecal incontinence

Study or sub-category	Treatment n/N	Control n/N	Weight %	OR (random) 95% CI
Klamt (Anaesth)	4/12	0/12	16.38	13.24 [0.63, 279.16]
Lauretti (RA-2)	3/20	2/20	41.82	1.59 [0.24, 10.70]
Lauretti (A & A -a)	0/10	0/10		Not estimable
Lauretti (A & A -c)	1/10	0/10	13.82	3.32 [0.12, 91.60]
Lauretti (A & A -b)	0/10	0/10		Not estimable
Almeida (AGYc)	0/12	0/12		Not estimable
Almeida (AGYb)	1/12	0/12	13.99	3.26 [0.12, 88.35]
Almeida (AGYa)	1/12	0/12	13.99	3.26 [0.12, 88.35]
Total (95% CI)	98	98	100.00	3.04 [0.89, 10.45]

Total events: 10 (Treatment), 2 (Control)
Test for heterogeneity: Chi^2=1.37, df=4 (P=0.85), I^2=0%
Test for overall effect: Z=1.77 (P=0.08)

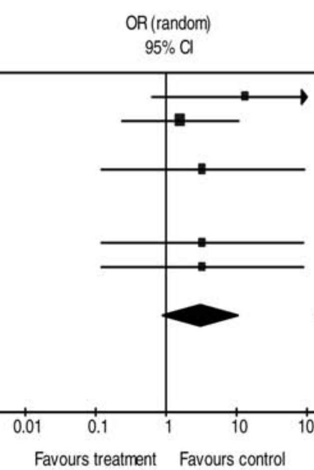


FIGURE 4: Forest plot showing the incidence of faecal incontinence after treatment with intrathecal neostigmine.

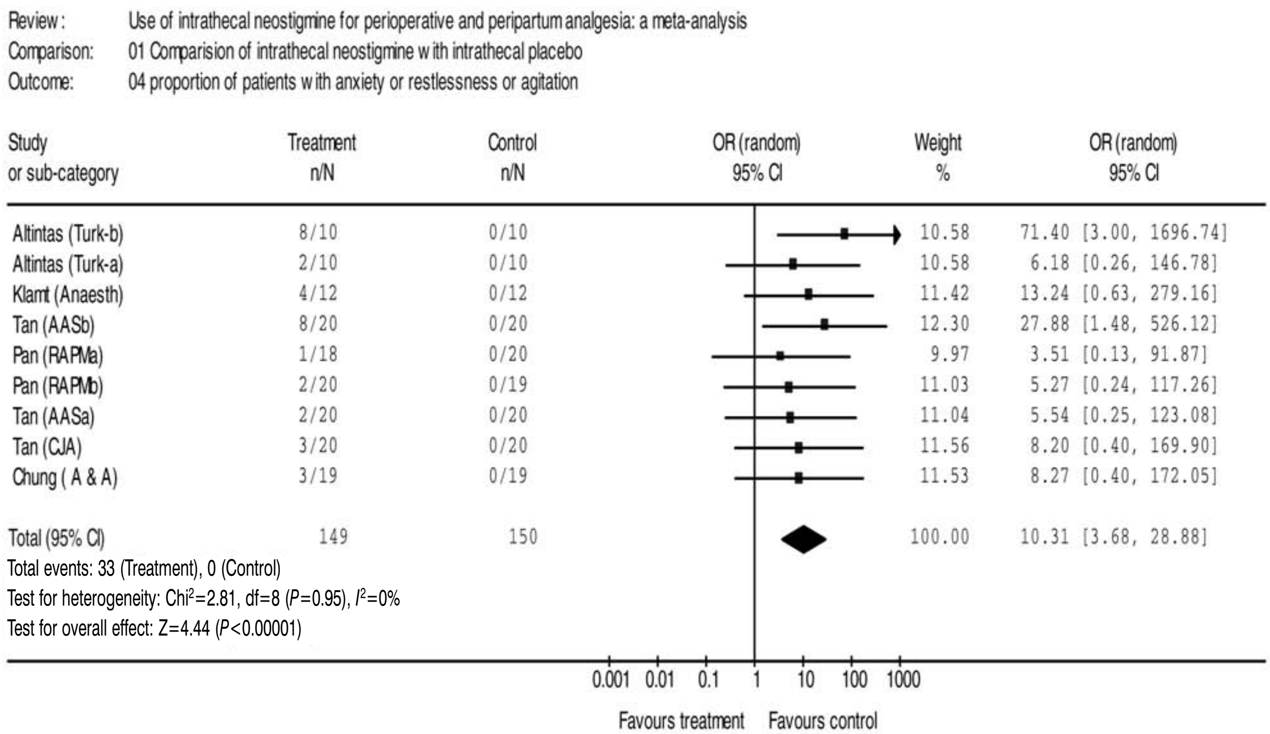


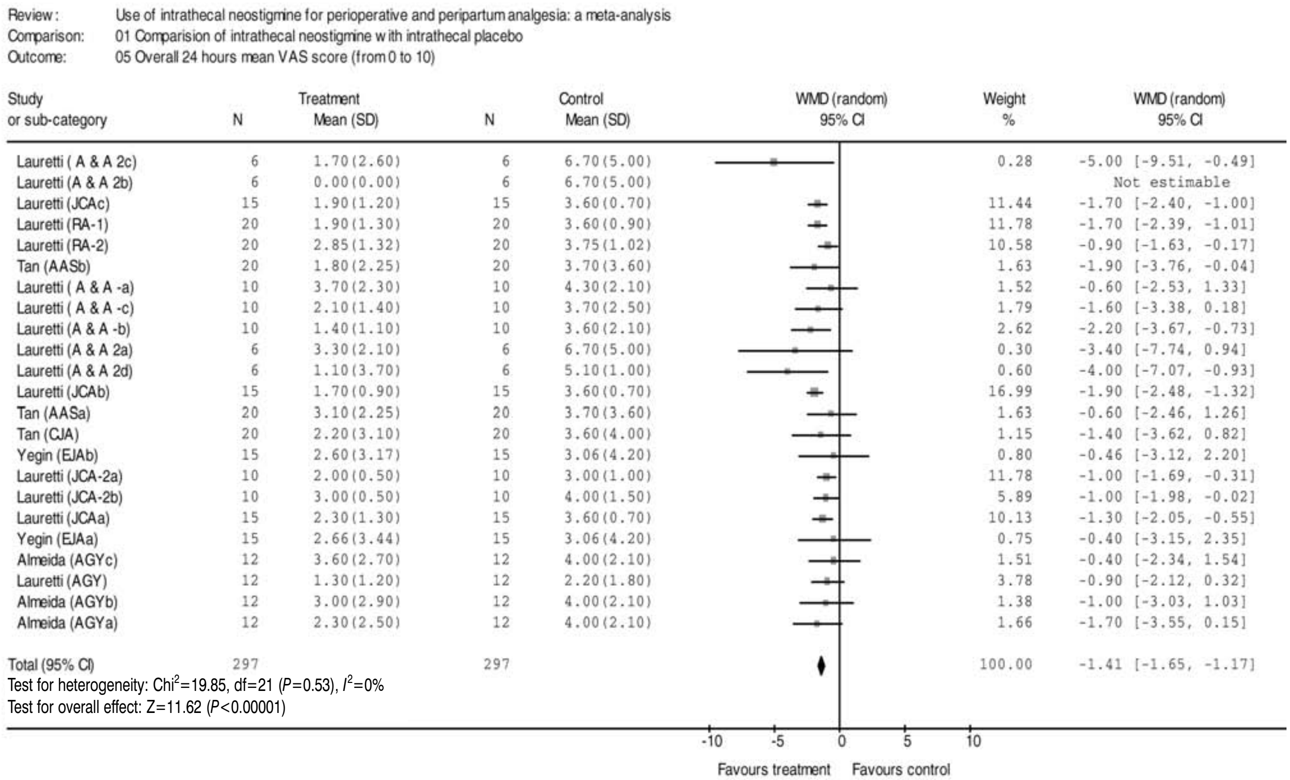
FIGURE 5: Forest plot showing the incidence of anxiety, restlessness or agitation after treatment with intrathecal neostigmine.

caesarean delivery or labour analgesia showed a similar trend in the results. Intrathecal neostigmine increased the incidence of nausea and vomiting (OR 16.1, 95% CI: 7.1 to 36.8; $P<0.00001$, $I^2=0\%$), and anxiety, agitation, or restlessness (OR 5.5, 95% CI: 0.9 to 33.3; $P=0.07$, $I^2=0\%$). It delayed the time of first request for rescue analgesia (179 min, 95% CI: 55 to 303; $P=0.005$, $I^2=98.5\%$), but did not increase the duration of motor blockade (54 min, 95% CI: –16 to 123; $P=0.13$, $I^2=73.2\%$) and did not reduce the total amount of ephedrine required to maintain haemodynamic stability (–0.4 mg, 95% CI: –5.2 to 5.9; $P=0.9$, $I^2=55.9\%$).

Sensitivity analysis by excluding studies with unclear allocation concealment did not change the direction and the magnitude of the results significantly. After excluding 9 studies with unclear allocation concealment, the odds ratio of the incidence of nausea and vomiting was 3.1 (95% CI: 1.9-5.0; $P<0.0001$, $I^2=13.5\%$), bradycardia was 2.8 (95% CI: 1.2-6.7; $P=0.04$, $I^2=0\%$), faecal incontinence was 2.1 (95% CI: 0.5 to 9.3; $P=0.3$, $I^2=0\%$), and anxiety or restlessness or agitation was 8.6 (95% CI: 2.2 to 24.3; $P=0.02$, $I^2=0\%$) after intrathecal neostigmine. The overall 24-hour VAS pain score was –1.4 (95% CI: –1.7 to –1.1; $P<0.0001$, $I^2=13.9\%$), the time of first request for analgesia 200 min (95% CI: 145 to 256; $P<0.0001$, $I^2=96.5\%$), the total number of NSAID injections was –0.8 (95% CI: –1.2 to 0.4; $P=0.0003$, $I^2=90.8\%$), the total amount of ephedrine required was 0.5 mg (95% CI: –2.0 to 1.0; $P=0.4$, $I^2=0\%$), and the increase in duration of motor block was 1.6 min (95% CI: –5.7 to 8.8; $P=0.93$, $I^2=0\%$) after intrathecal neostigmine. There was a small publication bias in this meta-analysis as demonstrated by the funnel plot (Figure 9), and small studies with severe nausea and vomiting after intrathecal neostigmine seemed less likely to be published. No pharmacoeconomic analysis was done in any of the included studies.

DISCUSSION

There was a good consistency in most outcomes of the studies included in this meta-analysis. Heterogeneity was observed only in two continuous outcome variables, the total number of rescue NSAID injections required in 24 hours and the time to request first rescue analgesia. By examining the mean difference between treatment and control group of each study down the forest plots, we could not observe any obvious dose dependent relationship in these two continuous outcome variables. The differences in intrathecal neostigmine dose are therefore unlikely to account for the heterogeneity. However, there was

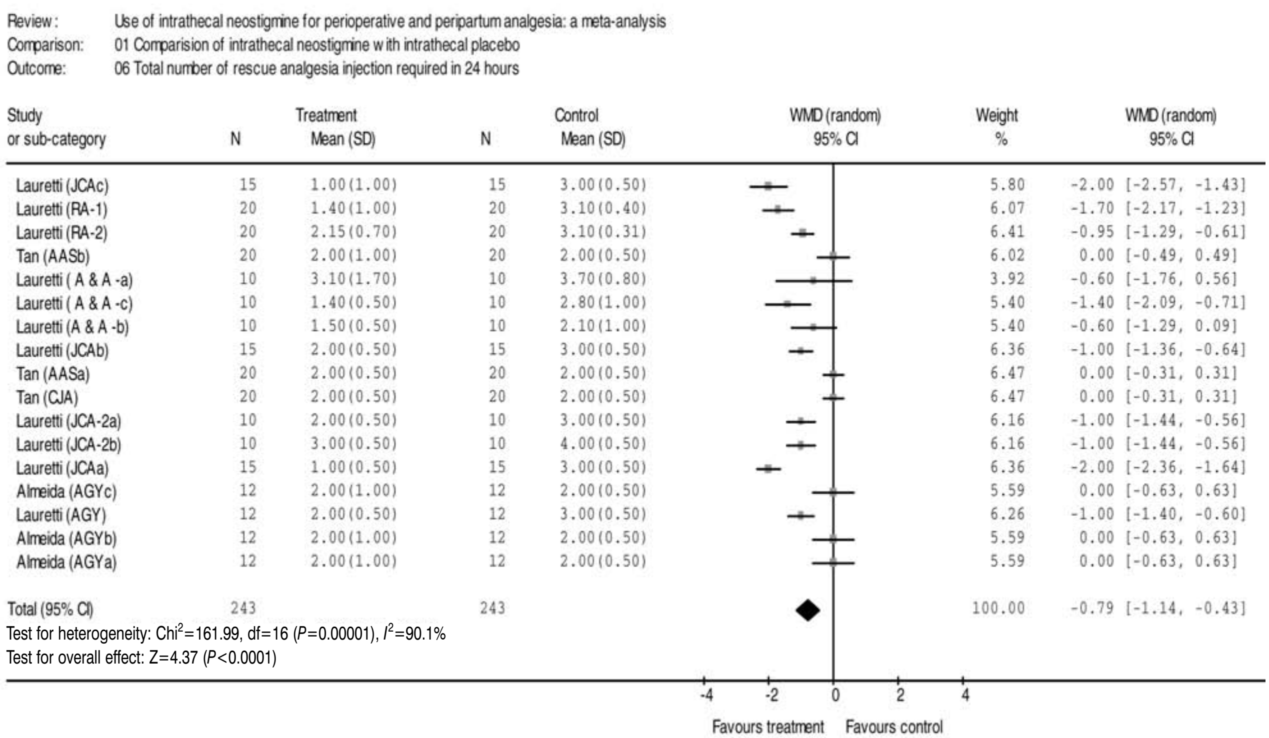


Review: Use of intrathecal neostigmine for perioperative and peripartum analgesia: a meta-analysis
Comparison: 01 Comparision of intrathecal neostigmine with intrathecal placebo
Outcome: 05 Overall 24 hours mean VAS score (from 0 to 10)

Study or sub-category	Treatment N	Treatment Mean (SD)	Control N	Control Mean (SD)	Weight %	WMD (random) 95% CI
Lauretti (A & A 2c)	6	1.70(2.60)	6	6.70(5.00)	0.28	-5.00 [-9.51, -0.49]
Lauretti (A & A 2b)	6	0.00(0.00)	6	6.70(5.00)		Not estimable
Lauretti (JCAc)	15	1.90(1.20)	15	3.60(0.70)	11.44	-1.70 [-2.40, -1.00]
Lauretti (RA-1)	20	1.90(1.30)	20	3.60(0.90)	11.78	-1.70 [-2.39, -1.01]
Lauretti (RA-2)	20	2.85(1.32)	20	3.75(1.02)	10.58	-0.90 [-1.63, -0.17]
Tan (AASb)	20	1.80(2.25)	20	3.70(3.60)	1.63	-1.90 [-3.76, -0.04]
Lauretti (A & A -a)	10	3.70(2.30)	10	4.30(2.10)	1.52	-0.60 [-2.53, 1.33]
Lauretti (A & A -c)	10	2.10(1.40)	10	3.70(2.50)	1.79	-1.60 [-3.38, 0.18]
Lauretti (A & A -b)	10	1.40(1.10)	10	3.60(2.10)	2.62	-2.20 [-3.67, -0.73]
Lauretti (A & A 2a)	6	3.30(2.10)	6	6.70(5.00)	0.30	-3.40 [-7.74, 0.94]
Lauretti (A & A 2d)	6	1.10(3.70)	6	5.10(1.00)	0.60	-4.00 [-7.07, -0.93]
Lauretti (JCAb)	15	1.70(0.90)	15	3.60(0.70)	16.99	-1.90 [-2.48, -1.32]
Tan (AASa)	20	3.10(2.25)	20	3.70(3.60)	1.63	-0.60 [-2.46, 1.26]
Tan (CJA)	20	2.20(3.10)	20	3.60(4.00)	1.15	-1.40 [-3.62, 0.82]
Yegin (EJAb)	15	2.60(3.17)	15	3.06(4.20)	0.80	-0.46 [-3.12, 2.20]
Lauretti (JCA-2a)	10	2.00(0.50)	10	3.00(1.00)	11.78	-1.00 [-1.69, -0.31]
Lauretti (JCA-2b)	10	3.00(0.50)	10	4.00(1.50)	5.89	-1.00 [-1.98, -0.02]
Lauretti (JCAa)	15	2.30(1.30)	15	3.60(0.70)	10.13	-1.30 [-2.05, -0.55]
Yegin (EJAa)	15	2.66(3.44)	15	3.06(4.20)	0.75	-0.40 [-3.15, 2.35]
Almeida (AGYc)	12	3.60(2.70)	12	4.00(2.10)	1.51	-0.40 [-2.34, 1.54]
Lauretti (AGY)	12	1.30(1.20)	12	2.20(1.80)	3.78	-0.90 [-2.12, 0.32]
Almeida (AGYb)	12	3.00(2.90)	12	4.00(2.10)	1.38	-1.00 [-3.03, 1.03]
Almeida (AGYa)	12	2.30(2.50)	12	4.00(2.10)	1.66	-1.70 [-3.55, 0.15]
Total (95% CI)	297		297		100.00	-1.41 [-1.65, -1.17]

Test for heterogeneity: $Chi^2=19.85$, df=21 ($P=0.53$), $I^2=0\%$
Test for overall effect: Z=11.62 ($P<0.00001$)

FIGURE 6: Forest plot showing the weight-mean-difference in 24 hours overall VAS pain score after treatment with intrathecal neostigmine. VAS=visual analog scale.

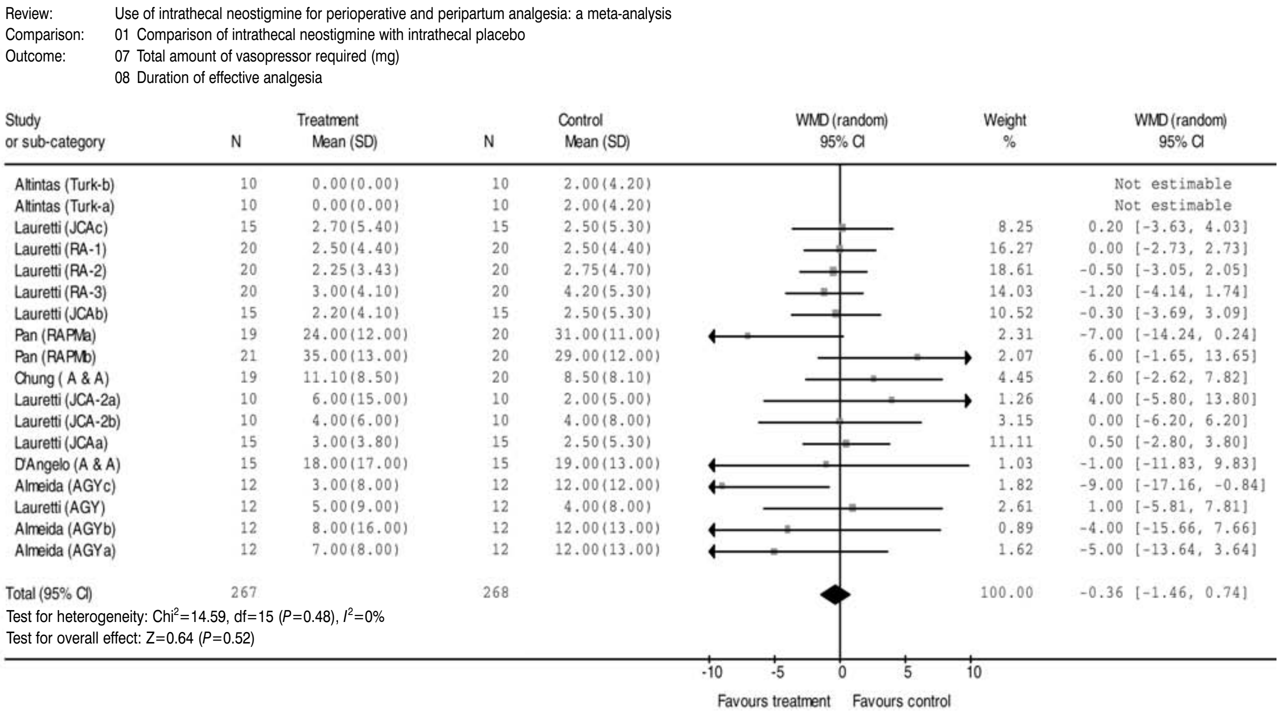


Review: Use of intrathecal neostigmine for perioperative and peripartum analgesia: a meta-analysis
Comparison: 01 Comparision of intrathecal neostigmine with intrathecal placebo
Outcome: 06 Total number of rescue analgesia injection required in 24 hours

Study or sub-category	Treatment N	Treatment Mean (SD)	Control N	Control Mean (SD)	Weight %	WMD (random) 95% CI
Lauretti (JCAc)	15	1.00(1.00)	15	3.00(0.50)	5.80	-2.00 [-2.57, -1.43]
Lauretti (RA-1)	20	1.40(1.00)	20	3.10(0.40)	6.07	-1.70 [-2.17, -1.23]
Lauretti (RA-2)	20	2.15(0.70)	20	3.10(0.31)	6.41	-0.95 [-1.29, -0.61]
Tan (AASb)	20	2.00(1.00)	20	2.00(0.50)	6.02	0.00 [-0.49, 0.49]
Lauretti (A & A -a)	10	3.10(1.70)	10	3.70(0.80)	3.92	-0.60 [-1.76, 0.56]
Lauretti (A & A -c)	10	1.40(0.50)	10	2.80(1.00)	5.40	-1.40 [-2.09, -0.71]
Lauretti (A & A -b)	10	1.50(0.50)	10	2.10(1.00)	5.40	-0.60 [-1.29, 0.09]
Lauretti (JCAb)	15	2.00(0.50)	15	3.00(0.50)	6.36	-1.00 [-1.36, -0.64]
Tan (AASa)	20	2.00(0.50)	20	2.00(0.50)	6.47	0.00 [-0.31, 0.31]
Tan (CJA)	20	2.00(0.50)	20	2.00(0.50)	6.47	0.00 [-0.31, 0.31]
Lauretti (JCA-2a)	10	2.00(0.50)	10	3.00(0.50)	6.16	-1.00 [-1.44, -0.56]
Lauretti (JCA-2b)	10	3.00(0.50)	10	4.00(0.50)	6.16	-1.00 [-1.44, -0.56]
Lauretti (JCAa)	15	1.00(0.50)	15	3.00(0.50)	6.36	-2.00 [-2.36, -1.64]
Almeida (AGYc)	12	2.00(1.00)	12	2.00(0.50)	5.59	0.00 [-0.63, 0.63]
Lauretti (AGY)	12	2.00(0.50)	12	3.00(0.50)	6.26	-1.00 [-1.40, -0.60]
Almeida (AGYb)	12	2.00(1.00)	12	2.00(0.50)	5.59	0.00 [-0.63, 0.63]
Almeida (AGYa)	12	2.00(1.00)	12	2.00(0.50)	5.59	0.00 [-0.63, 0.63]
Total (95% CI)	243		243		100.00	-0.79 [-1.14, -0.43]

Test for heterogeneity: $Chi^2=161.99$, df=16 ($P=0.00001$), $I^2=90.1\%$
Test for overall effect: Z=4.37 ($P<0.0001$)

FIGURE 7: Forest plot showing the weight-mean-difference in the total number of rescue NSAID injection required in 24 hours after intrathecal neostigmine. NSAID=nonsteroidal anti-inflammatory drug.



Review: Use of intrathecal neostigmine for perioperative and peripartum analgesia: a meta-analysis
Comparison: 01 Comparison of intrathecal neostigmine with intrathecal placebo
Outcome: 07 Total amount of vasopressor required (mg)
08 Duration of effective analgesia

Study or sub-category	N	Treatment Mean (SD)	N	Control Mean (SD)	WMD (random) 95% CI	Weight %	WMD (random) 95% CI
Altintas (Turk-b)	10	0.00(0.00)	10	2.00(4.20)			Not estimable
Altintas (Turk-a)	10	0.00(0.00)	10	2.00(4.20)			Not estimable
Lauretti (JCAc)	15	2.70(5.40)	15	2.50(5.30)		8.25	0.20 [-3.63, 4.03]
Lauretti (RA-1)	20	2.50(4.40)	20	2.50(4.40)		16.27	0.00 [-2.73, 2.73]
Lauretti (RA-2)	20	2.25(3.43)	20	2.75(4.70)		18.61	-0.50 [-3.05, 2.05]
Lauretti (RA-3)	20	3.00(4.10)	20	4.20(5.30)		14.03	-1.20 [-4.14, 1.74]
Lauretti (JCAb)	15	2.20(4.10)	15	2.50(5.30)		10.52	-0.30 [-3.69, 3.09]
Pan (RAPMa)	19	24.00(12.00)	20	31.00(11.00)		2.31	-7.00 [-14.24, 0.24]
Pan (RAPMb)	21	35.00(13.00)	20	29.00(12.00)		2.07	6.00 [-1.65, 13.65]
Chung (A & A)	19	11.10(8.50)	20	8.50(8.10)		4.45	2.60 [-2.62, 7.82]
Lauretti (JCA-2a)	10	6.00(15.00)	10	2.00(5.00)		1.26	4.00 [-5.80, 13.80]
Lauretti (JCA-2b)	10	4.00(6.00)	10	4.00(8.00)		3.15	0.00 [-6.20, 6.20]
Lauretti (JCAa)	15	3.00(3.80)	15	2.50(5.30)		11.11	0.50 [-2.80, 3.80]
D'Angelo (A & A)	15	18.00(17.00)	15	19.00(13.00)		1.03	-1.00 [-11.83, 9.83]
Almeida (AGYc)	12	3.00(8.00)	12	12.00(12.00)		1.82	-9.00 [-17.16, -0.84]
Lauretti (AGY)	12	5.00(9.00)	12	4.00(8.00)		2.61	1.00 [-5.81, 7.81]
Almeida (AGYb)	12	8.00(16.00)	12	12.00(13.00)		0.89	-4.00 [-15.66, 7.66]
Almeida (AGYa)	12	7.00(8.00)	12	12.00(13.00)		1.62	-5.00 [-13.64, 3.64]
Total (95% CI)	267		268			100.00	-0.36 [-1.46, 0.74]

Test for heterogeneity: Chi^2=14.59, df=15 (P=0.48), I^2=0%
Test for overall effect: Z=0.64 (P=0.52)

FIGURE 8: Forest plot showing the weight-mean-difference in the duration of effective analgesia (time to first request for rescue analgesia).

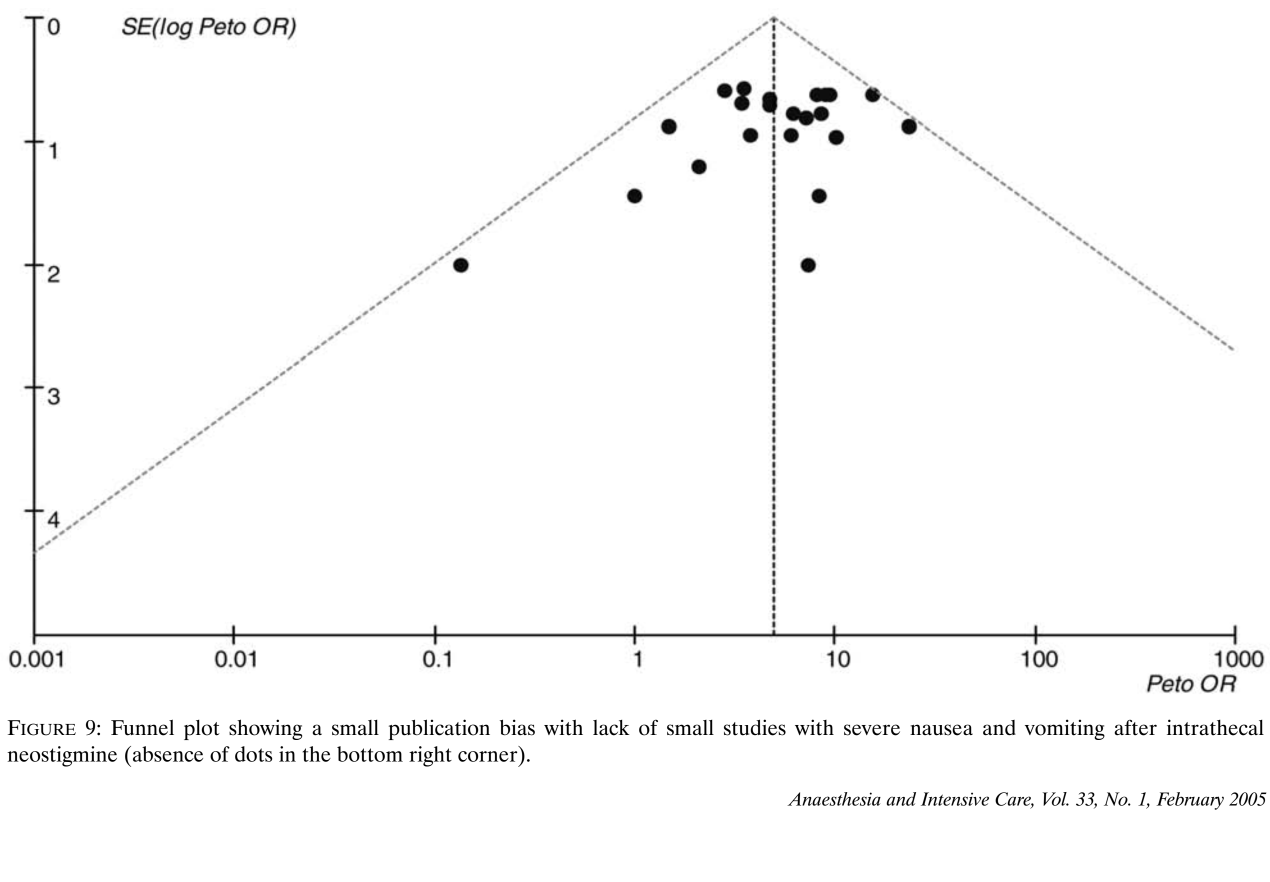


FIGURE 9: Funnel plot showing a small publication bias with lack of small studies with severe nausea and vomiting after intrathecal neostigmine (absence of dots in the bottom right corner).

significant variability, with large standard deviations in the results of each individual study in these two continuous outcome variables. Wide individual variations in the pain threshold and hence requirement for supplementary analgesia within the first 24 hours of surgery appears the most likely explanation. Also, the variety of surgical procedures in the analysis could be reflected in this heterogeneity.

Intrathecal neostigmine has been shown to be more effective in improving analgesia in the acute postoperative period[35], and thus we excluded volunteer studies from this meta-analysis. Nevertheless, the improvement in analgesia after adding intrathecal neostigmine to other spinal medications in the perioperative and peripartum settings was very small. While intrathecal neostigmine has been shown to have additive analgesic effect with NSAID in animal studies[36], our results did not demonstrate a dramatic reduction in NSAID consumption after intrathecal neostigmine. For the doses of intrathecal neostigmine (500 μg or less) examined in this meta-analysis, we also did not confirm the belief that intrathecal neostigmine improves haemodynamic stability when used with either intrathecal local anaesthetic or clonidine[2-4,37]. On the other hand, the increase in frequency of side-effects after intrathecal neostigmine was very significant. Many of these side-effects did not seem to be dose dependent and occurred even when the dose of intrathecal neostigmine was as low as 5 to 10 μg[9,16,17,19]. Furthermore, nausea and vomiting induced by intrathecal neostigmine was reported to be severe, repetitive, prolonged, and resistant to prevention or treatment by antiemetic drugs (such as metoclopramide, promethazine, ondansetron, droperidol, and dexamethasone)[9,17,24,26,34,38]. With the large sample size of this meta-analysis, we also noted some less well described side-effects of intrathecal neostigmine. Even low doses[26] appeared to increase the risk of agitation, restlessness and faecal incontinence, the latter being disturbing if the operative field was contaminated. Other unpleasant side-effects such as sweating above the level of the sensory block and intensive salivation were also observed in a few studies[18,23,29,33].

No pharmacoeconomic analysis was performed in any of the studies included in this meta-analysis. Improvement in perioperative analgesia by intrathecal neostigmine may reduce the requirement for rescue analgesics and thus reduce drug costs. However, side-effects of intrathecal neostigmine such as severe nausea and vomiting may require multiple antiemetic drugs and treatments, and delay recovery room discharge[9]. Based on the frequency and severity of side-effects associated with intrathecal neostigmine and the relatively minor improvements in analgesia, intrathecal neostigmine is unlikely to be cost-effective as an adjunct to other spinal medications. In contrast to intrathecal neostigmine, epidural neostigmine has been shown to improve labour analgesia and postoperative analgesia without many side-effects[39,40] and the use of epidural neostigmine in other clinical scenarios deserves further investigation.

Meta-analysis, as with any overview, is prone to bias. In order to avoid selection bias, we used three databases for the literature search and included studies written in different languages. Sensitivity analysis by excluding studies with unclear allocation concealment also did not change the direction and the magnitude of the results significantly. We thus believe that this meta-analysis provides a reliable overall picture of the effectiveness and side-effect profile of intrathecal neostigmine in acute pain settings.

In conclusion, adding intrathecal neostigmine to other spinal medications improves perioperative and peripartum analgesia only marginally when compared with placebo. It is associated with significant side-effects such as nausea and vomiting, bradycardia requiring atropine treatment, and anxiety, agitation or restlessness. The significant increase in side-effects outweighs the minor improvements in analgesia achieved by adding intrathecal neostigmine to other spinal medications. The routine use of intrathecal neostigmine as an adjunct to other spinal medications, to improve analgesia in the perioperative and peripartum setting, is not recommended.

REFERENCES

1. Kaksh TL, Dirksen R, Harty GJ. Antinociceptive effects of intrathecally injected cholinomimetic drugs in the rat and cat. Eur J Pharmacol 1985; 117:81-88.
2. Carp H, Jayaram A, Morrow D. Intrathecal cholinergic agonists lessen bupivacaine spinal-block-induced hypotension. Anesth Analg 1994; 79:112-116.
3. Hood DD, Eisenach JC, Tong C, Tommasi E, Yaksh TL. Cardiorespiratory and spinal cord blood flow effects of intrathecal neostigmine methylsulfate, clonidine, and their combination in sheep. Anesthesiology 1995; 82:428-435.
4. Rose G, Xu ZM, Tong CY, Eisenach JC. Spinal neostigmine diminishes, but does not abolish, hypotension from spinal bupivacaine in sheep. Anesth Analg 1996; 83:1041-1045.
5. Hood DD, Eisenach JC, Tuttle R. Phase I safety assessment of intrathecal neostigmine in humans. Anesthesiology 1995; 82:331-343.
6. Yaksh TL, Grafe MR, Malkmus S, Rathbun ML, Eisenach JC. Studies on safety of chronically administered intrathecal neostigmine methylsulfate in rats and dogs. Anesthesiology 1995; 82:412-427.
7. Walker SM, Goudas LC, Cousins MJ, Carr DB. Combination spinal analgesic chemotherapy: a systematic review. Anesth Analg 2002; 95:674-715.

8. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. BMJ 2003; 327:557-560.
9. Liu SS, Hodgson PS, Moore JM, Trautman WJ, Burkhead DL. Dose-response effects of spinal neostigmine added to bupivacaine spinal anesthesia in volunteers. Anesthesiology 1999; 90:710-717.
10. Hood DD, Mallak KA, Eisenach JC, Tong C. Interaction between intrathecal neostigmine and epidural clonidine in human volunteers. Anesthesiology 1996; 85:315-325.
11. Hood DD, Mallak KA, James RL, Tuttle R, Eisenach JC. Enhancement of analgesia from systemic opioid in humans by spinal cholinesterase inhibition. J Pharmacol Exp Ther 1997; 282:86-92.
12. Klamt JG, Garcia LV, Prado WA. Analgesic and adverse effects of a low dose of intrathecally administered hyperbaric neostigmine alone or combined with morphine in patients submitted to spinal anaesthesia: pilot studies. Anaesthesia 1999; 54:27-31.
13. Nelson KE, D'Angelo R, Foss ML, Meister GC, Hood DD, Eisenach JC. Intrathecal neostigmine and sufentanil for early labor analgesia. Anesthesiology 1999; 91:1293-1298.
14. Lauretti GR, Lima IC. The effects of intrathecal neostigmine on somatic and visceral pain: improvement by association with a peripheral anticholinergic. Anesth Analg 1996; 82:617-620.
15. Krukowski JA, Hood DD, Eisenach JC, Mallak KA, Parker RL. Intrathecal neostigmine for post-cesarean section analgesia: dose response. Anesth Analg 1997; 84:1269-1275.
16. Owen MD, Özsaraç Ö, Şahin Ş, Uçkunkaya N, Kaplan N, Mağunaci I. Low-dose clonidine and neostigmine prolong the duration of intrathecal bupivacaine-fentanyl for labor analgesia. Anesthesiology 2000; 92:361-366.
17. D'Angelo R, Dean LS, Meister GC, Nelson KE. Neostigmine combined with bupivacaine, clonidine, and sufentanil for spinal labor analgesia. Anesth Analg 2001; 93:1560-1564.
18. Lauretti GR, Reis MP, Prado WA, Klamt JG. Dose-response study of intrathecal morphine versus intrathecal neostigmine, their combination, or placebo for postoperative analgesia in patients undergoing anterior and posterior vaginoplasty. Anesth Analg 1996; 82:1182-1187.
19. Almeida RA, Lauretti GR, Mattos AL. Antinociceptive effect of low-dose intrathecal neostigmine combined with intrathecal morphone following gynecologic surgery. Anesthesiology 2003; 98:495-498.
20. Lauretti GR, Mattos AL, Reis MP, Pereira NL. Combined intrathecal fentanyl and neostigmine: therapy for postoperative abdominal hysterectomy pain relief. J Clin Anesth 1998; 10:291-296.
21. Pan PM, Huang CT, Wei TT, Mok MS. Enhancement of analgesic effect of intrathecal neostigmine and clonidine on bupivacaine spinal anesthesia. Reg Anesth Pain Med 1998; 23:49-56.
22. Lauretti GR, Azevedo VM. Intravenous ketamine or fentanyl prolongs postoperative analgesia after intrathecal neostigmine. Anesth Analg 1996; 83:766-770.
23. Altintas F, Tunali Y, Utku T, Pozkurt P, Kaya G, Köse Y. Spinal Anestezide Bupivakain+Neostigmin Uygulamasi. Türk Anest Rean Mecmuast 1997; 25:313-317.
24. Yegin A, Yilmaz M, Karsli B, Erman M. Analgesic effects of intrathecal neostigmine in perianal surgery. Eur J Anaesthesiol 2003; 20:404-408.
25. Lauretti GR, Mattos AL, Reis MP, Prado WA. Intrathecal neostigmine for postoperative analgesia after orthopedic surgery. J Clin Anesth 1997; 9:473-477.
26. Tan PH, Kuo JH, Liu K, Hung CC, Tsai TC, Deng TY. Efficacy of intrathecal neostigmine for the relief of postinguinal herniorrhaphy pain. Acta Anaesthesiol Scand 2000; 44:1056-1060.
27. Lauretti GR, Hood DD, Eisenach JC, Pfeifer BL. A multicenter study of intrathecal neostigmine for analgesia following vaginal hysterectomy. Anesthesiology 1998; 89:913-918.
28. Lauretti GR, Reis MP. Subarachnoid neostigmine does not affect blood pressure or heart rate during bupivacaine spinal anesthesia. Reg Anesth 1996; 21:586-591.
29. Klamt JG, Slullitel A, Garcia IV, Prado WA. Postoperative analgesic effect of intrathecal neostigmine and its influence on spinal anaesthesia. Anaesthesia 1997; 52:547-551.
30. Lauretti GR, Oliveira AP, Juliao MC, Reis MP, Pereira NL. Transdermal nitroglycerine enhances spinal neostigmine postoperative analgesia following gynecological surgery. Anesthesiology. 2000; 93:943-946.
31. Lauretti GR, Mattos AL, Gomes JM, Pereira NL. Postoperative analgesia and antiemetic efficacy after intrathecal neostigmine in patients undergoing abdominal hysterectomy during spinal anesthesia. Reg Anesth 1997; 22:527-533.
32. Chung CJ, Kim JS, Park HS, Chin YJ. The efficacy of intrathecal neostigmine, intrathecal morphine, and their combination for post-cesarean section analgesia. Anesth Analg 1998; 87:341-346.
33. Lauretti GR, Reis MP. Postoperative analgesia and antiemetic efficacy after subarachnoid neostigmine in orthopedic surgery. Reg Anesth 1997; 22:337-342.
34. Tan PH, Chia YY, Lo Y, Liu K, Yang LC, Lee TH. Intrathecal bupivacaine with morphine or neostigmine for postoperative analgesia after total knee replacement surgery. Can J Anesth 2001; 48:551-556.
35. Bouaziz H, Tong C, Eisenach JC. Postoperative analgesia from intrathecal neostigmine in sheep. Anesth Analg 1995; 80:1140-1144.
36. Miranda HF, Sierralta F, Pinardi G. Neostigmine interactions with nonsteroidal anti-inflammatory drugs. Br J Pharmacol 2002; 135:1591-1597.
37. Williams JS, Tong C, Eisenach JC. Neostigmine counteracts spinal clonidine-induced hypotension in sheep. Anesthesiology 1993; 78:301-307.
38. Tan PH, Liu K, Peng CH, Yang LC, Lin CR, Lu CY. The effect of dexamethasone on postoperative pain and emesis after intrathecal neostigmine. Anesth Analg 2001; 92:228-232.
39. Roelants F, Lavand'homme PM. Epidural neostigmine combined with sufentanil provides balanced and selective analgesia in early labor. Anesthesiology 2004; 101:439-444.
40. Omais M, Lauretti GR, Paccola CA. Epidural morphine and neostigmine for postoperative analgesia after orthopedic surgery. Anesth Analg 2002; 95:1698-1701.

Appendix 1: The details of the included studies

Study ID	Design	Participants	Interventions	Outcomes	Allocation concealment
Almeida (AGYa) (AGYb) (AGYc)	double-blind RCT	60 abdominal hysterectomy	IT morphine+ bupivacaine: 3 doses of neostigmine (1, 2.5, 5 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. 24-h VAS score 5. total number of rescue analgesics 6. amount of vasopressor (mg) 7. faecal incontinence	Adequate
Altintas (Turk-a) (Turk-b)	single-blind RCT	30 varico-ectomy or orchidectomy	IT bupivacaine neostigmine: 2 doses: 200 μg in (a) 500 μg in (b)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. amount of vasopressor (mg) 5. agitation / restlessness	Unclear
Chung (A & A)	unclear blinding RCT	80 elective caesarean delivery	IT bupivacaine± neostigmine: one dose (25 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. amount of vasopressor (mg) 5. total number of rescue analgesics 6. anxiety	Unclear
D'Angelo (A & A)	double-blind RCT	36 analgesia for active labour	IT sufentanil+ clonidine+ bupivacaine+ neostigmine: one dose (100 μg)	1. nausea & vomiting 2. bradycardia 3. effective analgesia (min) 4. amount of vasopressor (mg)	Adequate
Klamt (Anaesth)	double-blind RCT	24 anterior and posterior vaginoplasty	IT bupivacaine± neostigmine: one dose (100 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. faecal incontinence 5. restlessness	Unclear
Lauretti (A & A-a) (A & A-b) (A & A-c)	double-blind RCT	60 anterior and posterior vaginoplasty	IT bupivacaine: in (b)±IT fentanyl), in (c) ±ketamine), all with or without neostigmine: one dose (50 μg)	1 nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. overall 24-h VAS score 5. total number of rescue analgesics 6. faecal incontinence 7. bradycardia	Unclear
Lauretti (A & A 2a) (A & A 2b) (A & A 2c) (A & A 2d)	double-blind RCT	48 anterior and posterior vaginoplasty	In (a) GA+IT neostigmine (50 μg) In (b) GA+IT neostigmine 100 μg In (b) GA+IT neostigmine 100 μg In (c) GA+200 μg In (d) GA+IT morphine 50 μg+ IT neostigmine 50 μg	1. nausea & vomiting 2. effective analgesia (min) 3. 24-h overall VAS score 4. bradycardia 5. dizziness	Adequate

Appendix 1 cont.

Study ID	Design	Participants	Interventions	Outcomes	Allocation concealment
Lauretti (AGY)	double-blind RCT	48 vaginoplasty	IT bupivacaine± neostigmine: one dose (5 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. overall 24-h VAS score 5. total number of rescue analgesics 6. amount of vasopressor required (mg)	Adequate
Lauretti (AGY-2a) (AGY-2b) (AGY-2c)	double-blind RCT	92 vaginal hysterectomy	IT bupivacaine± neostigmine: 3 doses: 25 μg in (a), 50 μg in (b), 75 μg in (c)	1. nausea & vomiting 2. bradycadia 3. effective analgesia (min)	Adequate
Lauretti (JCA-2a) (JCA-2b)	double-blind RCT	50 abdominal hysterectomy	In (a) IT bupivacaine+ fentanyl± neostigmine (25 μg) In (b) IT bupivacaine± neostigmine (25 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. overall 24-h VAS score 5. total number of rescue analgesics 6. amount of vasopressor (mg) 7. bradycardia	Adequate
Lauretti (JCAa) (JCAb) (JCAc)	double-blind RCT	60 below knee surgery	IT bupivacaine± neostigmine: 3 doses: 25 μg in (a), 50 μg in (b), 100 μg in (c)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. overall 24-h VAS score 5. amount of vasopressor required (mg) 6. bradycardia	Adequate
Lauretti (RA-1)	double-blind RCT	100 tibial and ankle reconstruction	IT bupivacaine± neostigmine: one dose (100 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. overall 24-h VAS score 5. total number of rescue analgesics 6. amount of vasopressor (mg) 7. bradycardia	Unclear
Lauretti (RA-2)	double-blind RCT	100 abdominal hysterectomy	IT bupivacaine± neostigmine: one dose (100 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. total number of rescue analgesics 5. amount of vasopressor (mg) 6. faecal incontinence 7. bradycardia	Adequate
Lauretti (RA-3)	single-blind RCT	40 abdominal hysterectomy	IT neostigmine± neostigmine: one dose (75 μg)	1. nausea & vomiting 2. motor blockade (min) 3. bradycardia 4. amount of vaospressor required (mg)	

Appendix 1 cont.

Study ID	Design	Participants	Interventions	Outcomes	Allocation concealment
Owen (AGY)	double-blind RCT	50 analgesia for active labour	IT (bup+fentanyl +clonidine)± IT neostigmine one dose (10 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. bradycardia	Not used
Pan (RAPM-a) (RAPM-b)	double-blind RCT	80 elective caesarean delivery	IT bupivacaine± neostigmine: one dose (50 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. amount of vasopressor required (mg) 5. dizziness/anxiety/restlessness	Unclear
Tan (AASa) (AASb)	double-blind RCT	60 inguinal hernia operation	IT tetracaine± neostigmine: 2 doses: 50 μg in (a), 100 μg in (b)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. overall 24-h VAS score 5. total number of rescue analgesics 6. amount of vasopressor required (mg) 7. dizziness/anxiety	Adequate
Tan (CJA)	double-blind RCT	62 total knee replacement	IT bupivacaine± neostigmine: one dose (50 μg)	1. nausea & vomiting 2. effective analgesia (min) 3. duration of motor blockade (min) 4. overall 24-h VAS score 5. total number of rescue analgesics 6. anxiety/dizziness	Unclear
Yegin (EJAa) (EJBb)	double-blind RCT	45 perianal surgery	IT bupivacaine± neostigmine: 2 doses: 25 μg in (a), 50 μg in (b)	1. nausea & vomiting 2. effective analgesia (min) 3. motor blockade (min) 4. overall 24-h VAS score 5. amount of vasopressor required (mg) 6. bradycardia	Unclear